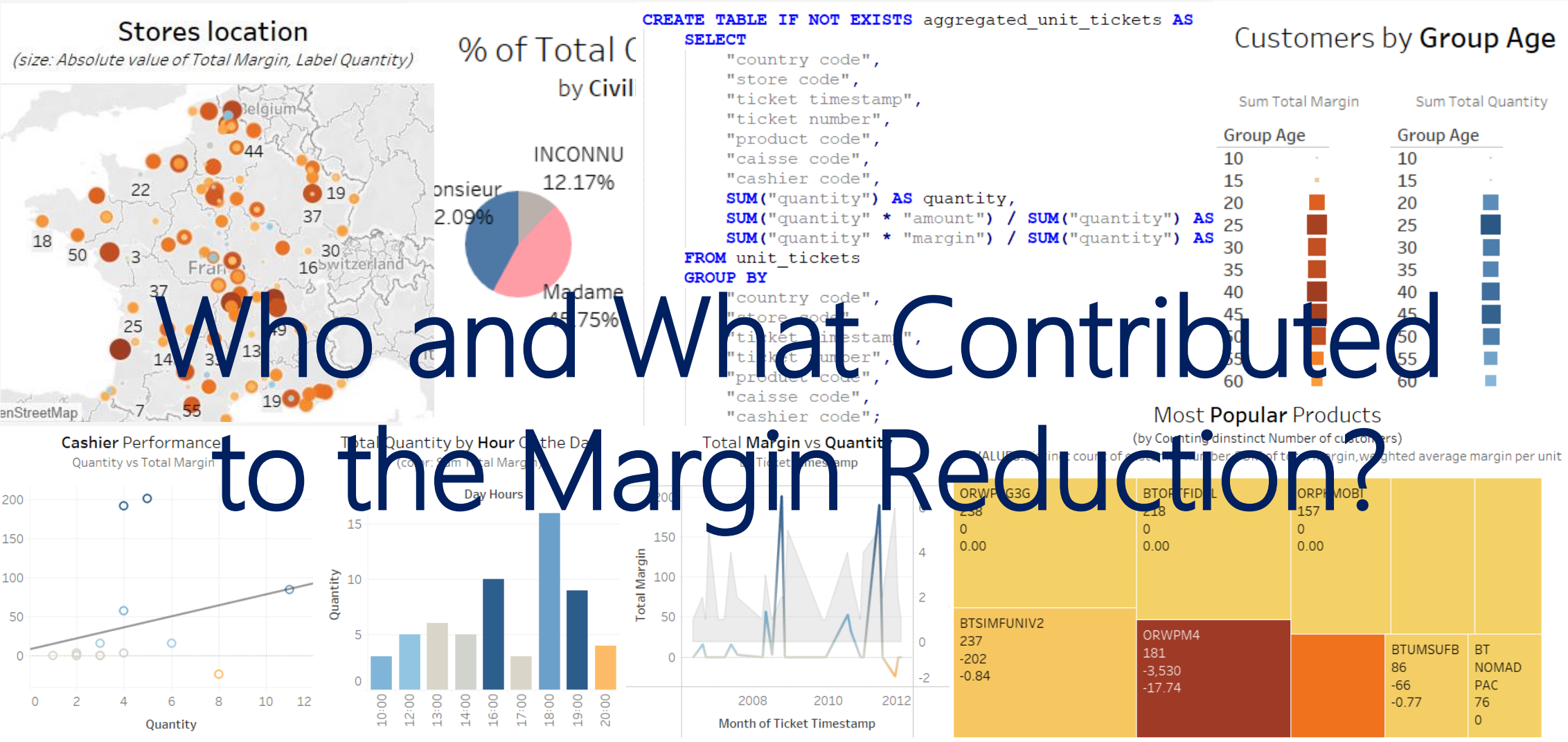


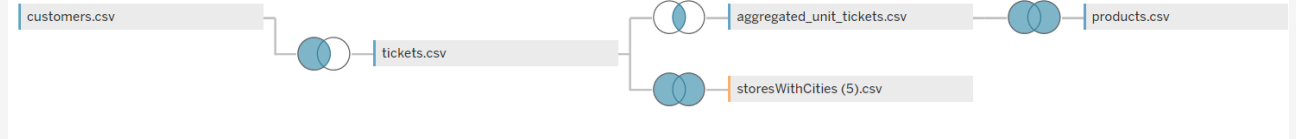


Who and What Contributed to the Margin Reduction?

Introduction

1 Join Reasons

customers.csv is made of 5 tables. ⓘ



2 Brief explanation of the modified **unit_tickets** table

3 Goal of the analysis and Introduction to the Dashboards

4 Indicators:

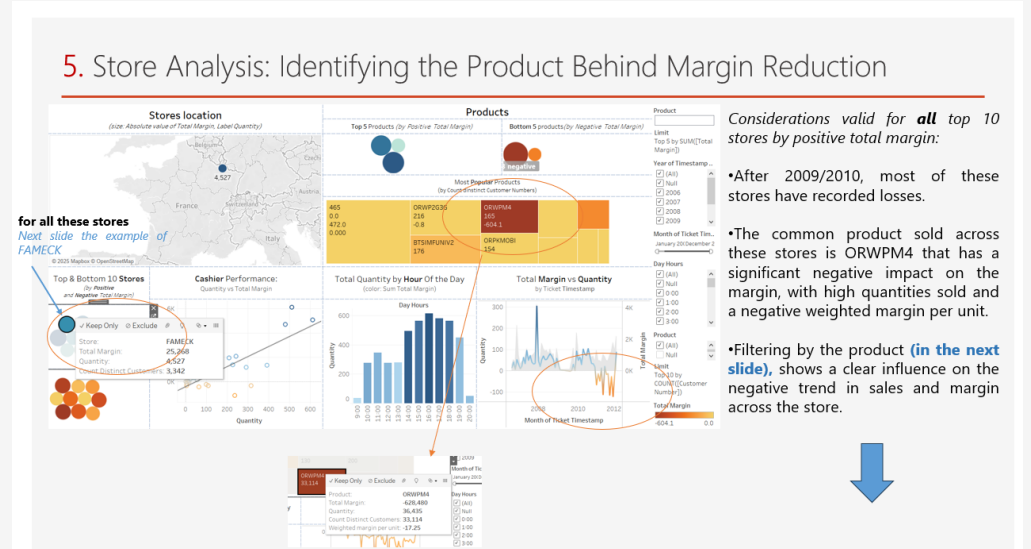
4.1. Key indicators

4.2. Stores and Products indicators

4.3. Customer indicators

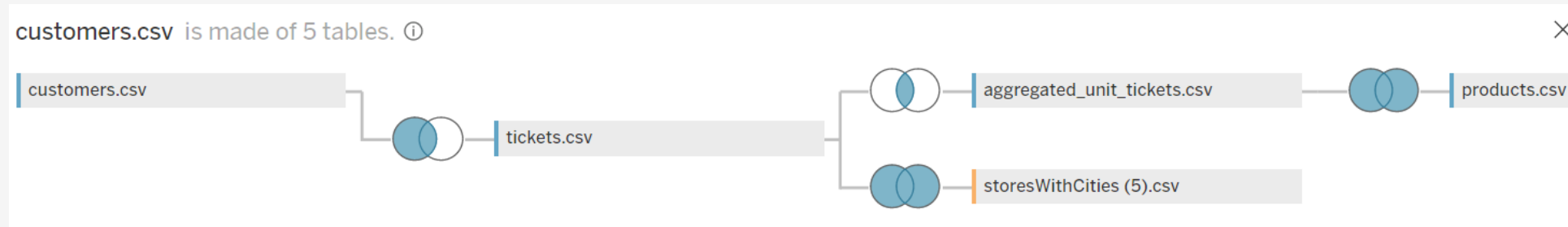
5 Interactive Storytelling->Identifying the Causes of Margin Reduction

6 Conclusion



1. Join Reasons

"FRA",1.903608000491E12,"Madame","ANNA","1938-06-30 00:00:00.000",9036,9036
"FRA",1.903608000492E12,"Monsieur","ALAIN","1969-03-15 00:00:00.000",9036,9036
"FRA",1.903608000493E12,"Madame","CATHERINE","2999-01-01 00:00:00.000",9036,9036
"FRA",1.903608000494E12,"Madame","MARIE-MADELEINE","1947-09-04 00:00:00.000",9036,9036
.....



- The analysis focuses on **customers**, **products** and **stores**
- **INNER JOIN:** Tickets not associated with unit_tickets and viceversa will not be analyzed
- **FULL OUTER JOINS** are used to ensure that all stores and all products are preserved, even if they were not sold or associated with a ticket

CUSTOMERS

"country code", "customer number", "civility", "first name", "birthday", "store code creation", "store code attachment"

PRODUCTS

"product code", "product"

STORES

"country code", "store code", "store"

TICKETS

"country code", "store code", "timestamp ticket", "ticket number", "customer number"

UNIT_TICKETS

"country code", "store code", "ticket timestamp", "ticket number", "line number", "product code", "caisse code", "cashier code", "quantity", "amount", "margin"

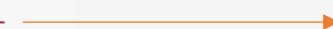
977256 TICKETS are not associated with any UNIT_TICKET. 0 TICKETS are not associated with any CUSTOMER 0 CUSTOMERS are not associated with any TICKET.

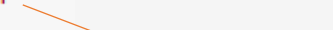
- Each customer is associated with at least one ticket, and vice versa
- Not all tickets are associated with unit_ticket

2. Reduce unit_tickets table

```
cursor.execute("""
    CREATE TABLE IF NOT EXISTS aggregated_unit_tickets AS
    SELECT
        "country code",
        "store code",
        "ticket timestamp",
        "ticket number",
        "product code",
        "caisse code",
        "cashier code",
        SUM("quantity") AS quantity,
        SUM("quantity" * "amount") / SUM("quantity") AS amount,
        SUM("quantity" * "margin") / SUM("quantity") AS margin
    FROM unit_tickets
    GROUP BY
        "country code",
        "store code",
        "ticket timestamp",
        "ticket number",
        "product code",
        "caisse code",
        "cashier code";
""")
```

- It aggregates the same product appearing multiple times in the same ticket
- Associate a weighted average to the amount and margin


$$\frac{\sum[\text{Amount}] * [\text{Quantity}]}{\sum[\text{Quantity}]}$$

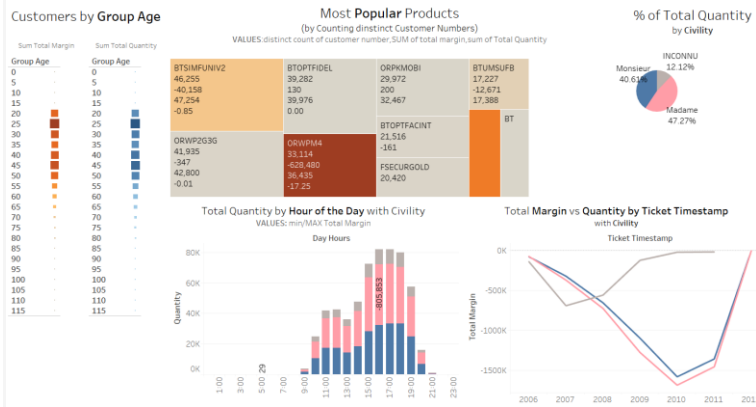

$$\frac{\sum[\text{Margin}] * [\text{Quantity}]}{\sum[\text{Quantity}]}$$

3. Goal

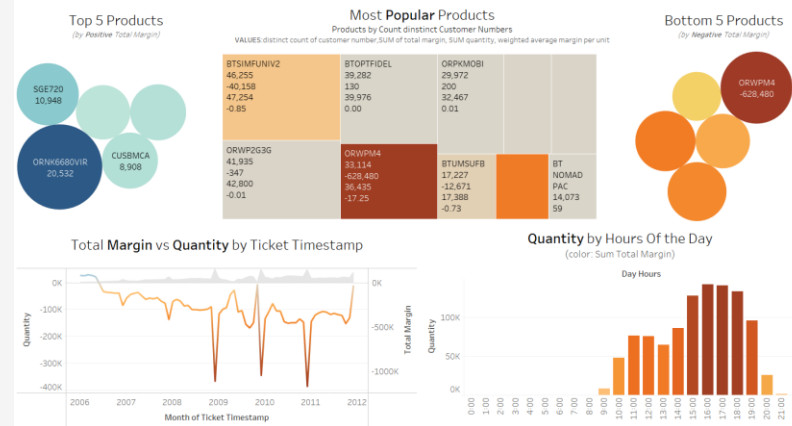
The analysis focuses on 3 aspects: **products**, **stores** and **customers**, aiming to identify how the sale of certain products has impacted the overall margin.

There are 3 dashboards, each dedicated to one of these elements.

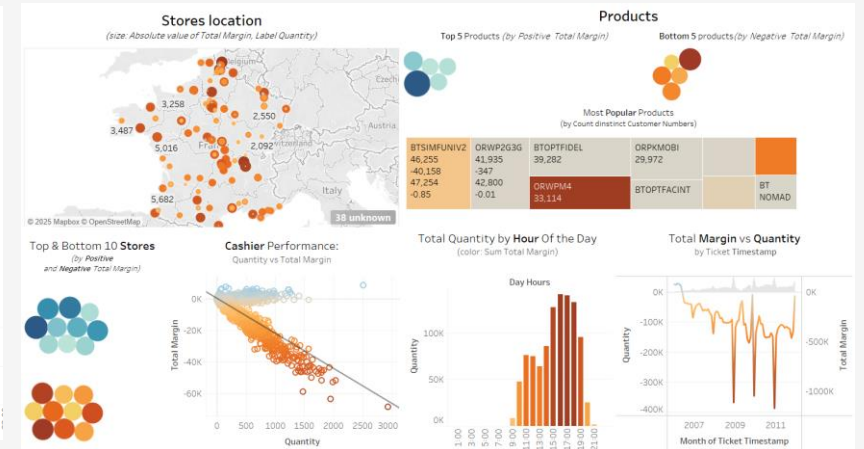
1 Customers



2 Products



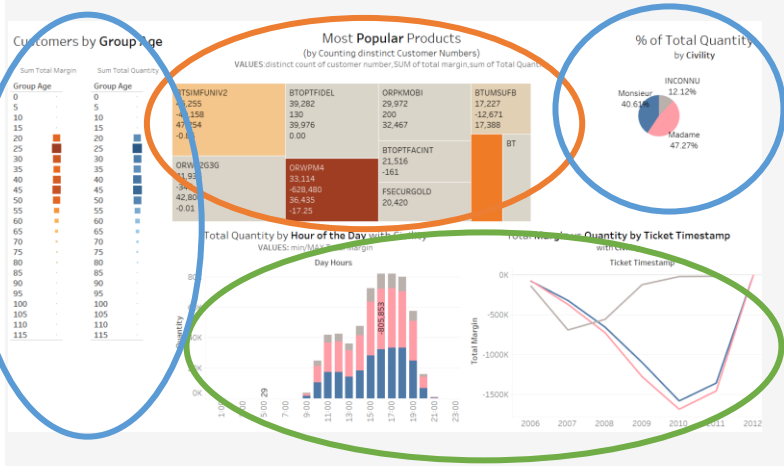
3 Stores



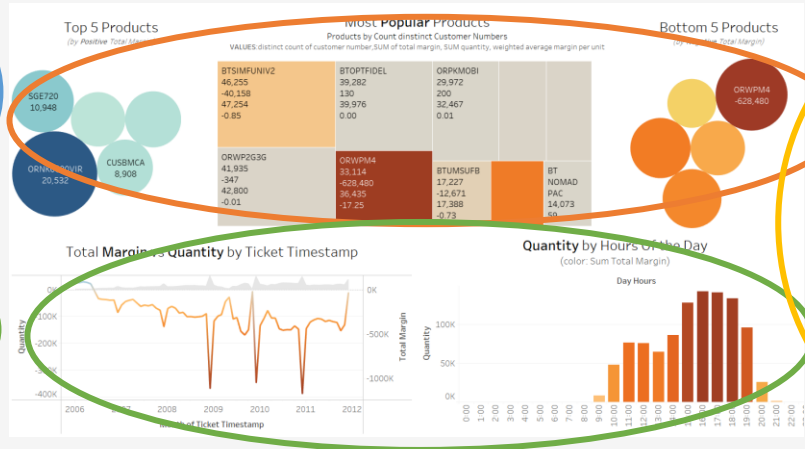
3. Dashboards

The indicators are divided into **key** indicators, **product** indicators, **customer** indicators, and **store** indicators.

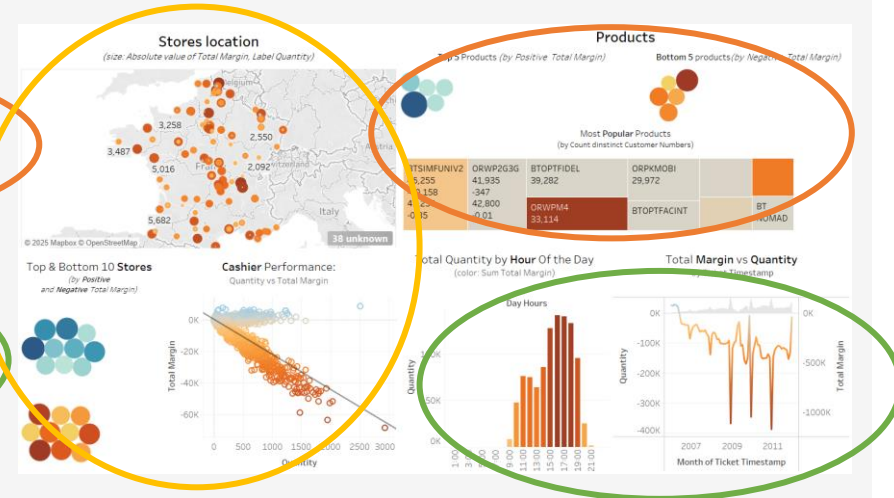
1 Customers



2 Products



3 Stores

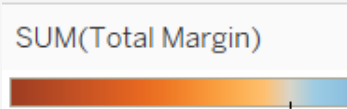


- The same **product** indicators and **key** indicators are present on all 3 dashboards
- The key indicators on the customer dashboard differ only in color (by Civility), but analyze the same aspects as on the other dashboards.

4. Introduction to the Indicators

All indicators in the 3 dashboards (*except 3 indicators for the analysis of the customers*) are colored by: **Total margin sum**

- **Blue** for **Positive** Total Margin
- **Orange** for **Negative** Total Margin



Why?

to visually highlight losses and gains

Next the explanation of the indicators:

1. Key indicators
2. Product and Store indicators
3. Customer Indicators

Followed by the analysis

4.1. Key Indicators

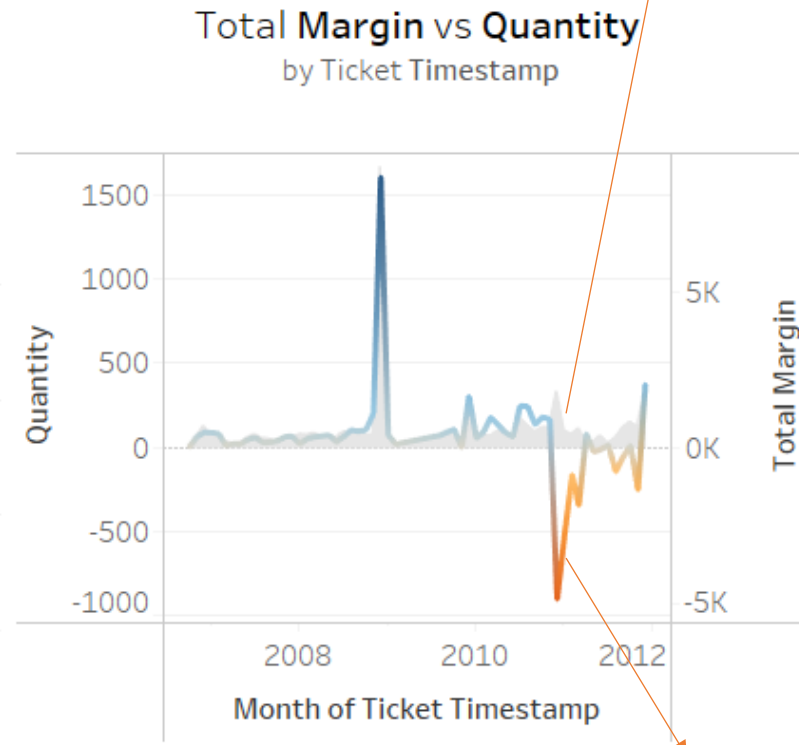
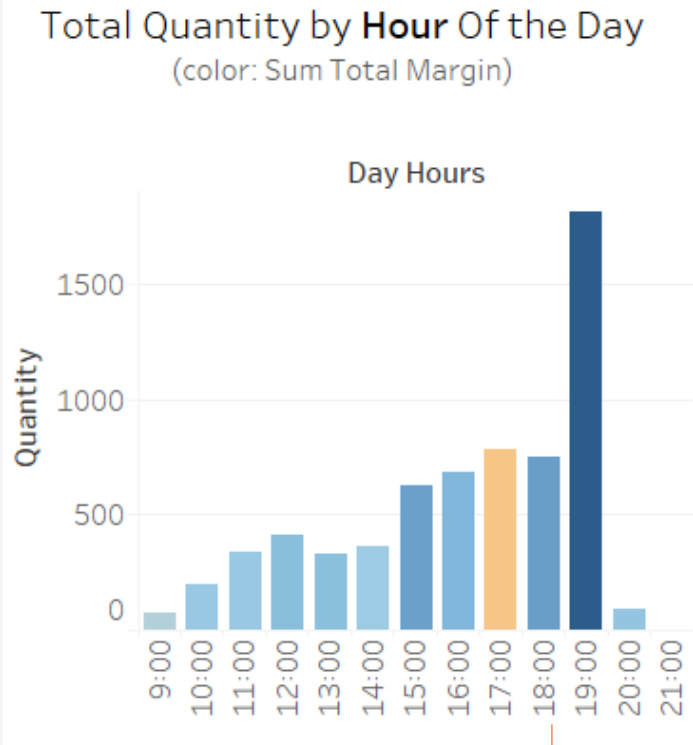
These 2 indicators are present on all **3** dashboards and allow for analyzing the margin and quantity sold across different **products**, **stores**, **age groups**, **civilities** and **cashiers**.

Filters that allow analyzing data by months of the year or by hours of the day

Month of Ticket Tim..
January 20(December 2

Day Hours

- ☒ (All)
- ☒ Null
- ☒ 9:00
- ☒ 10:00
- ☒ 11:00
- ☒ 12:00
- ☒ 13:00
- ☒ 14:00
- ☒ 15:00
- ☒ 16:00
- ☒ 17:00
- ☒ 18:00
- ☒ 19:00
- ☒ 20:00
- ☒ 21:00



Day Hours

customers.csv+ (Multiple Connections)

```
STR(DATEPART('hour', [Timestamp Ticket])) + ":00"
```

Why use the dual axis?

In some cases, the total margin is zero. It helps determine whether this is due to a lack of sales during that period or because there was no profit despite the sales

4.2. Products and Stores Indicators

4 Bubble charts:

- Size: Sum total margin

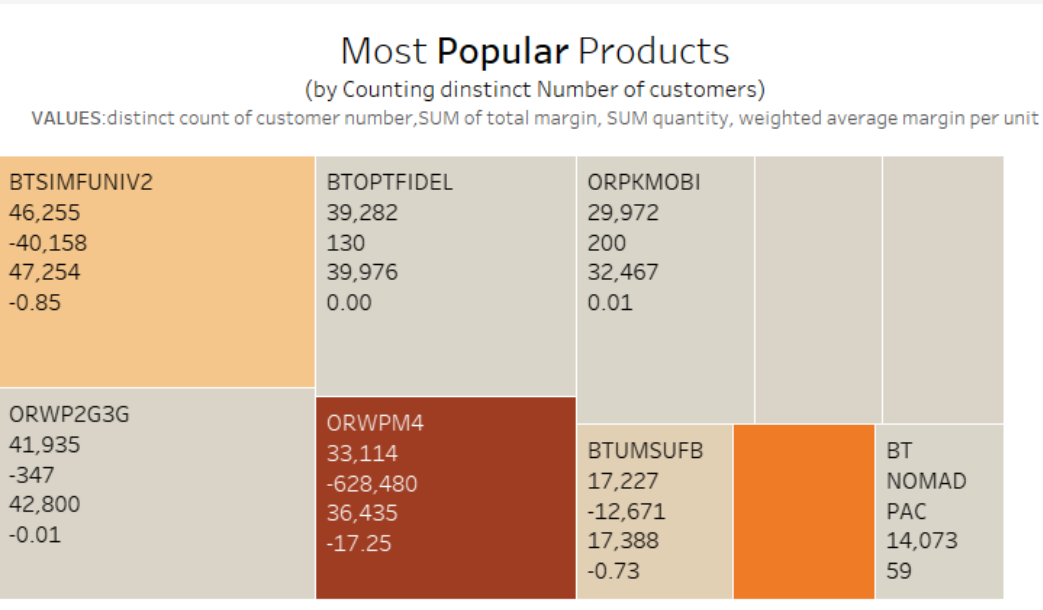
Used to analyze the **products** and **stores** that generated the highest positive margins and the lowest negative margins



1 Treemap:

- Size: Count Distinct Customer Number

Used to analyze the products purchased by the highest number of distinct customers, **“Popular products”**.



4.2. Product Indicator: Treemap

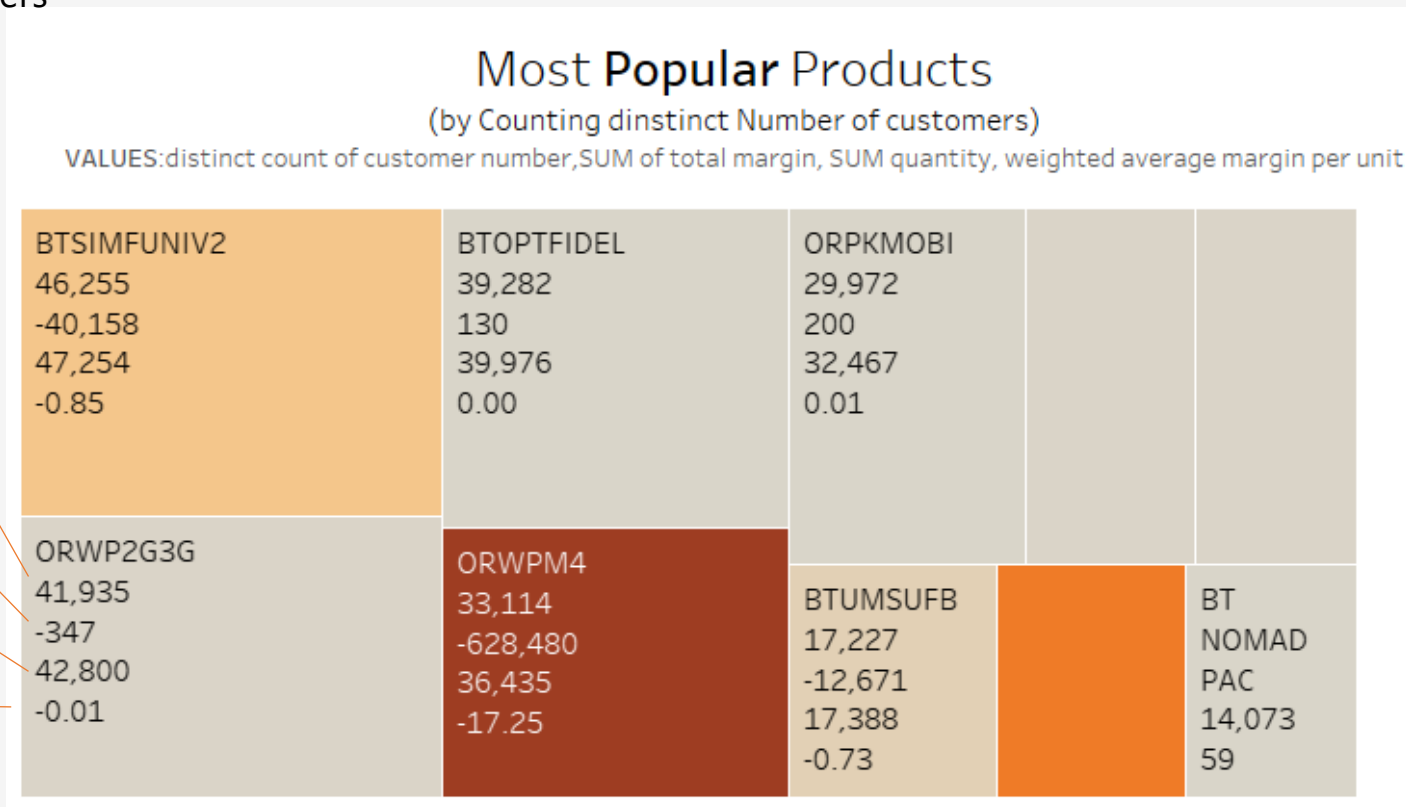
1. Count distinct Customer Numbers

2. Total Margin

3. Sum Quantity

4. Weighted margin per unit

$$\frac{\sum[\text{Margin}] * [\text{Quantity}]}{\sum[\text{Quantity}]}$$



Quantity: None
Quantity: 0.0
Quantity: 1.0
Quantity: 2.0
Quantity: 3.0
Quantity: 4.0
Quantity: 5.0
Quantity: 6.0
Quantity: 7.0
Quantity: 8.0
Quantity: 9.0
Quantity: 10.0
Quantity: 12.0
Quantity: 13.0
Quantity: 14.0
Quantity: 15.0
Quantity: 16.0
Quantity: 17.0
Quantity: 18.0
Quantity: 19.0
Quantity: 20.0
Quantity: 22.0
Quantity: 24.0
Quantity: 25.0
Quantity: 26.0
Quantity: 27.0
Quantity: 28.0
Quantity: 29.0
Quantity: 32.0
Quantity: 35.0
Quantity: 36.0
Quantity: 37.0
Quantity: 41.0
Quantity: 48.0
Quantity: 52.0
Quantity: 89.0
Quantity: 99.0

product quantities can vary, it allows for a precise evaluation of profitability

4.2. Store Indicators

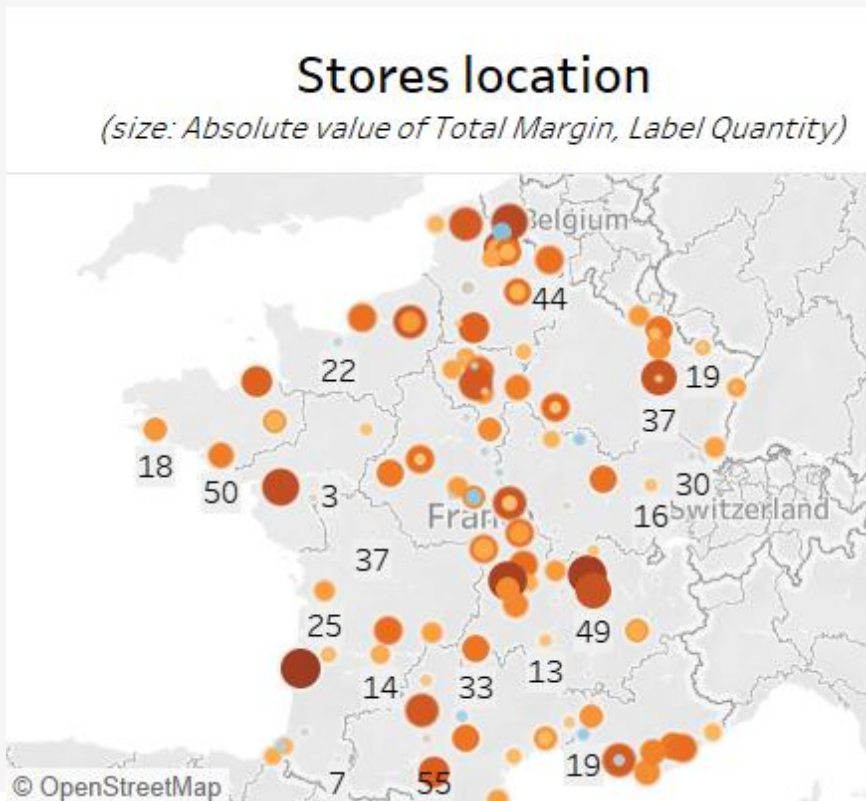
Map:

Size: absolute value of the total margin

Color: sum of the total margin.

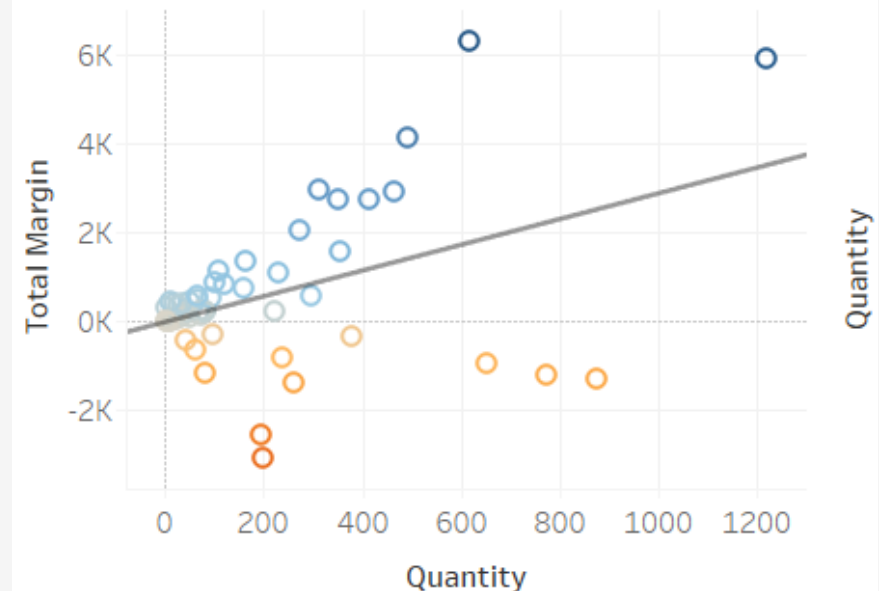
Label: sum of the quantity

Allows to analyze where the products have been sold and identifies areas with more negative margins



Cashier Performance:

Quantity vs Total Margin

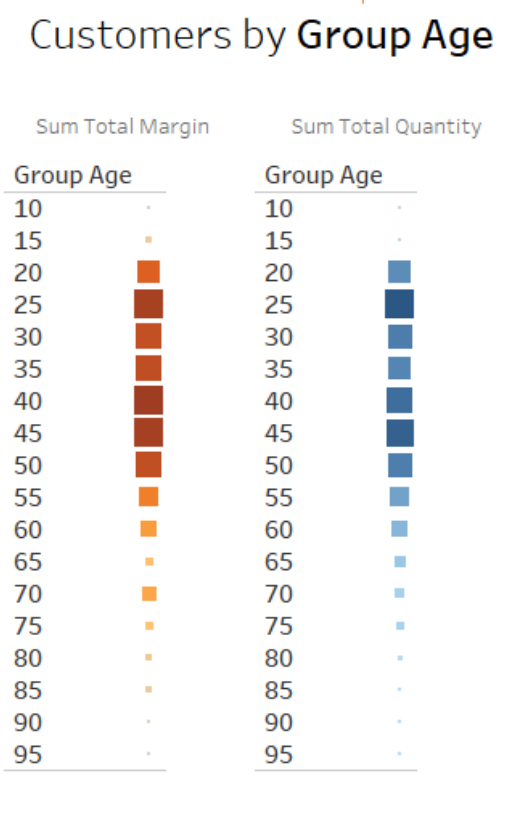


1. It allows identifying which cashiers generate the highest and lowest margin relative to the quantities sold.
2. When compared with the main indicator for "margin by hours of the day," it helps assess a specific cashier's performance throughout the day
3. It is possible to identify cashiers who have sold a specific product or work at a specific store by using the previous indicators as filters

4.3. Customer Indicators

Customer Age customers.csv+ (Multiple Connections) ×

FLOAT(DATEDIFF('year', [Birthday], DATE("2013-01-01")))|



- 1. Customer age calculated from January 2013, since the latest recorded sales are from the beginning of 2012
- 2. Also used as filter

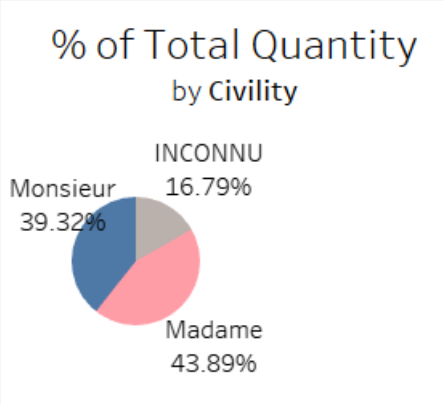
Age Parameter

5

☐ (All)

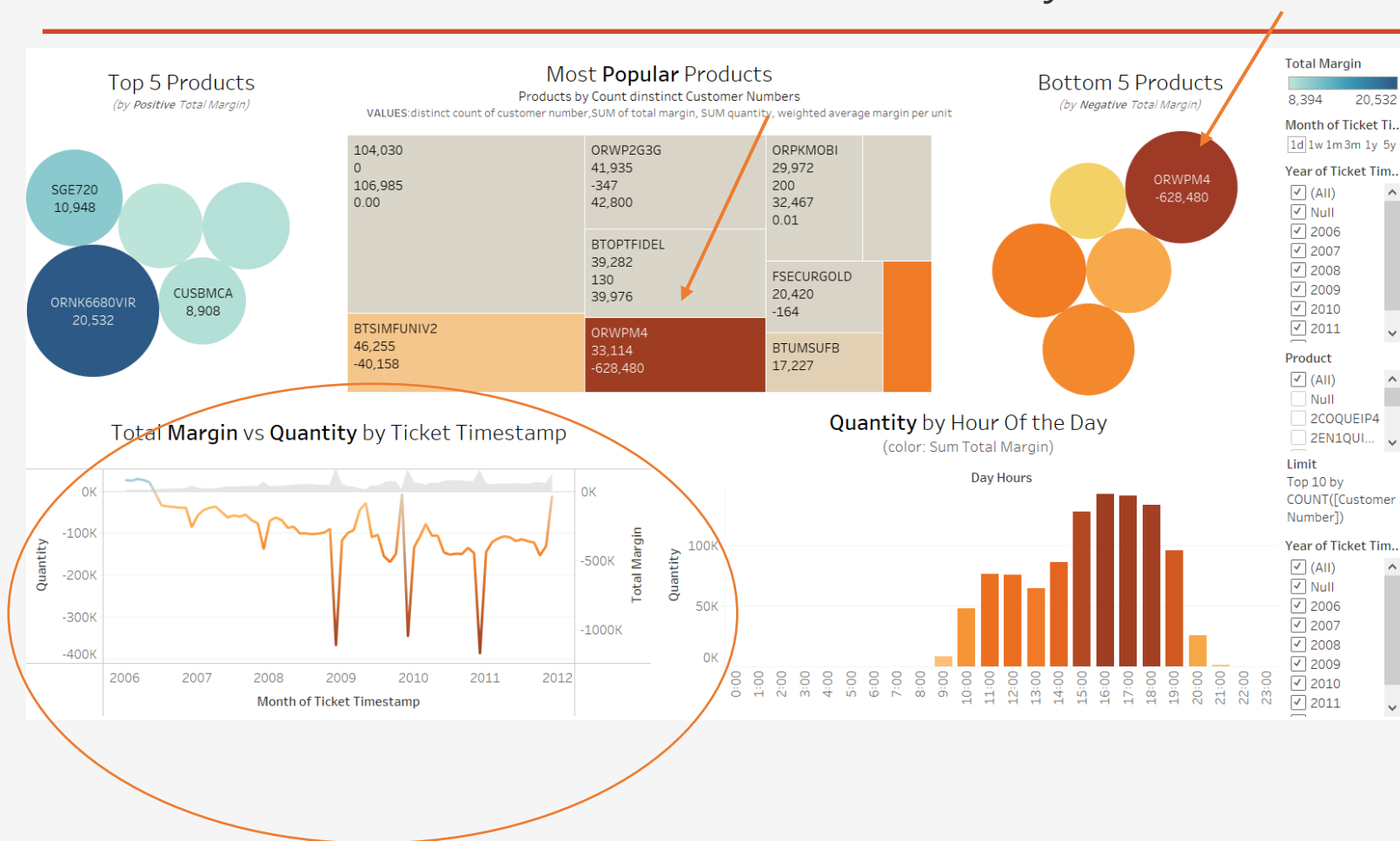
☒ Null

☐ 10



Pie chart :
shows the percentage of total quantity sold based on civility

5. General Consideration for the analysis



Considering all products, all customers, and all stores, the overall margin is negative.

Starting with a positive margin in 2006, it has shown negative peaks in subsequent years.

One of the product that have had the greatest impact is **ORWPM4**:

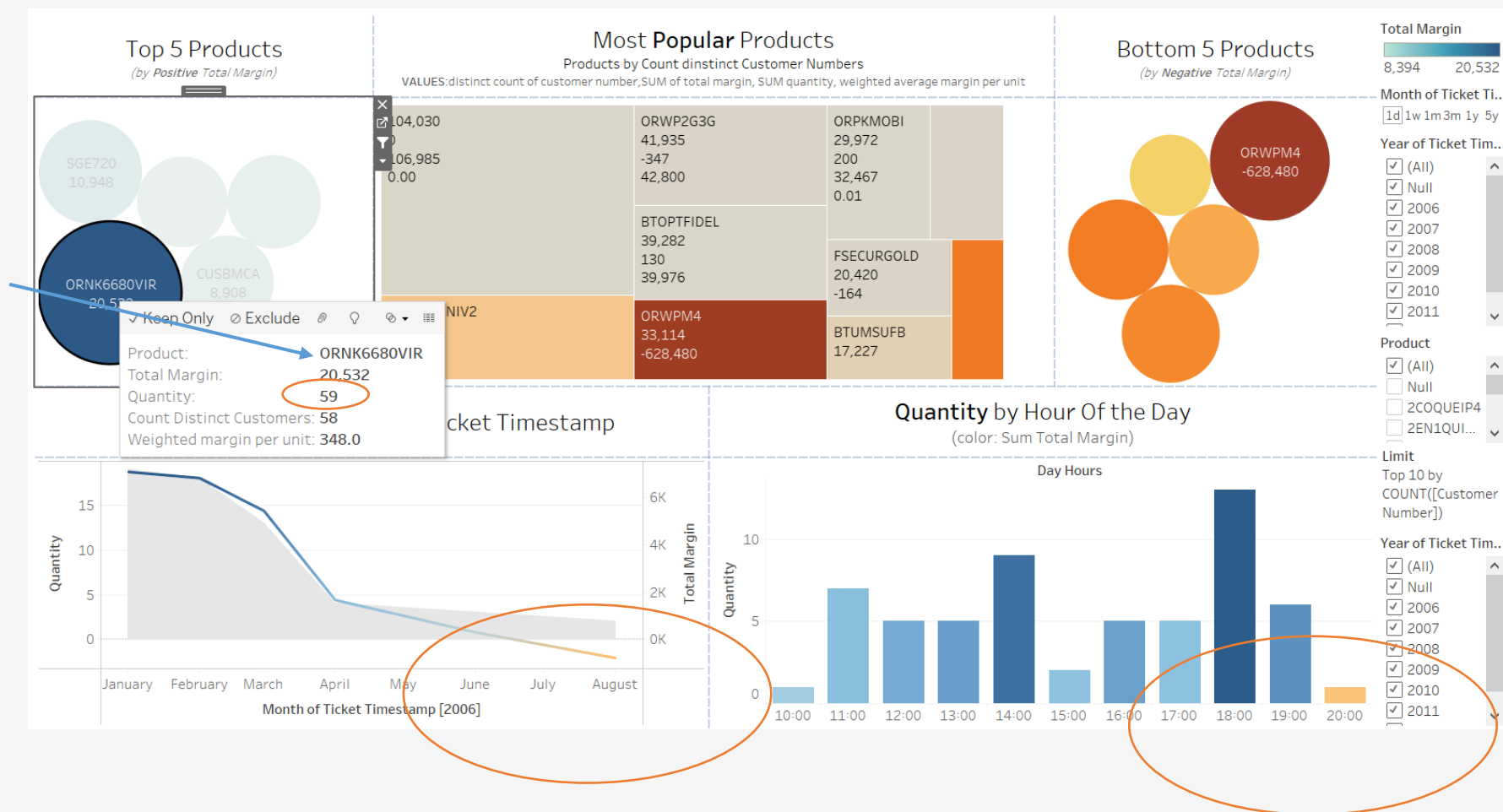
- Purchased by the majority of distinct customers (popular product)
- Sold by most stores (including those with the highest positive total margin)
- Has the most negative margin

In **contrast**, the products with the highest positive total margin:

- had sales limited to 2006.
- showed lower quantities sold compared to other products.

In the following slides, the behavior of the products with the highest margins is analyzed to understand how they have influenced the total margin. Subsequently, the product ORWPM4 is analyzed, along with its distribution across stores and the types of customers who purchased it.

5. Analysis of Products with the Highest Positive Margin



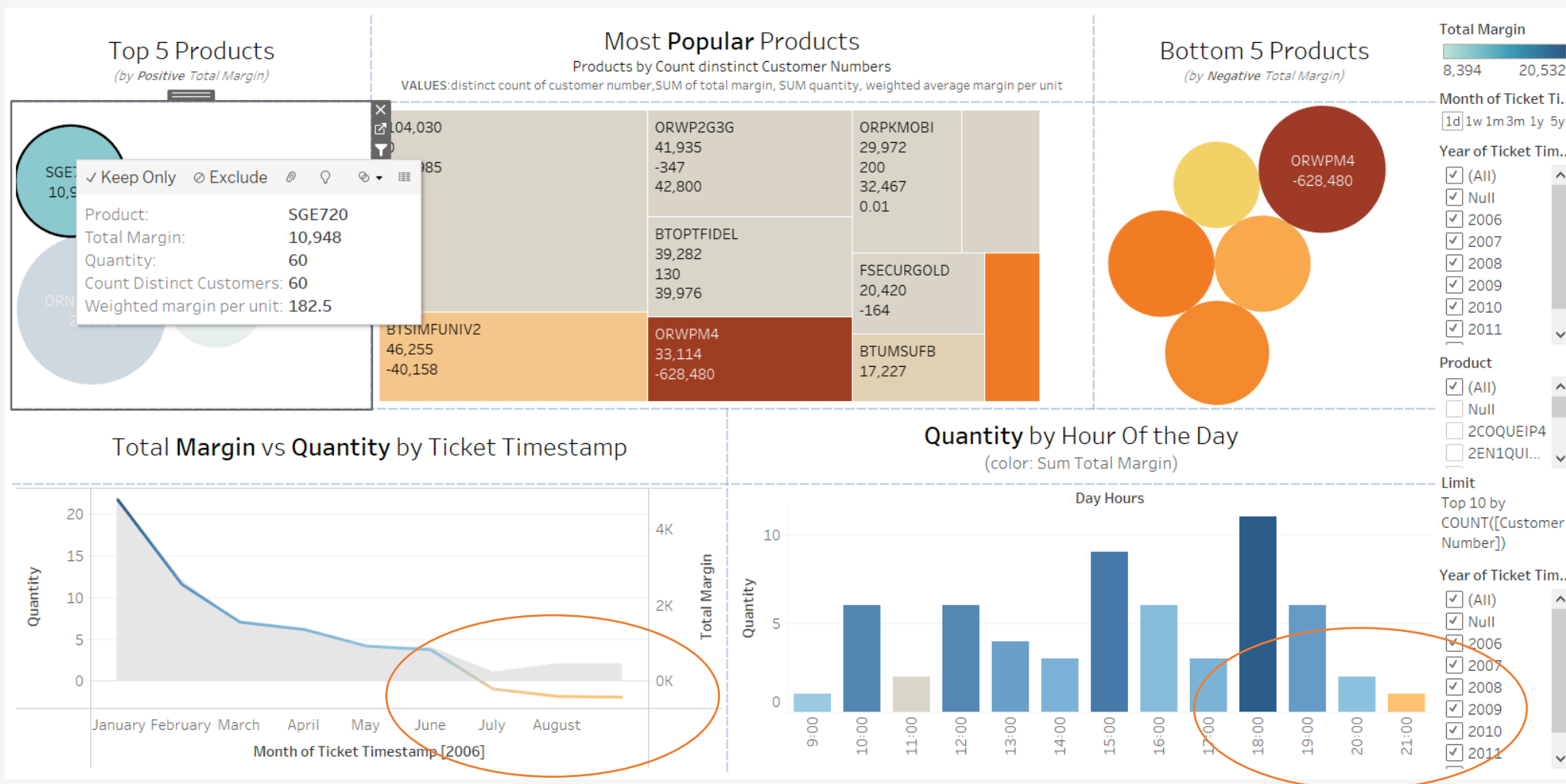
The two products with the highest Total Margin behave similarly:

1. After August/September 2006 they are no longer sold.
2. In late evening 20:00/21:00, the margin becomes negative
3. The quantities sold are significantly lower compared to the other products.
4. The margin decreases over time until it becomes negative
5. They have a high weighted margin per unit -> probably the reason why are in the top 5 by Total Margin

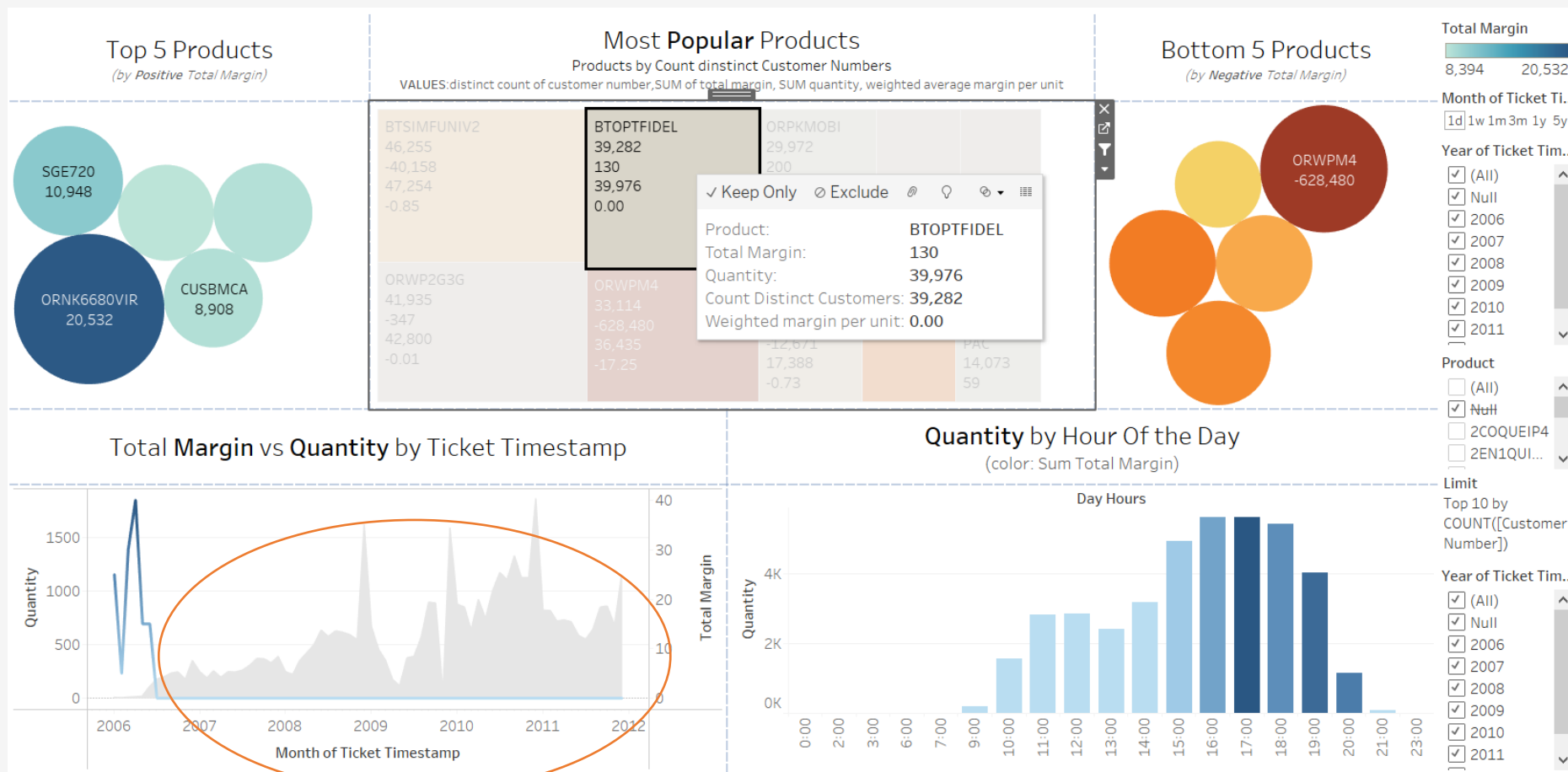
The product with the **second** highest total margin is in the next slide



5. Analysis of Products with the Highest Positive Margin



5. Analysis of the Most Popular Products Among Customers



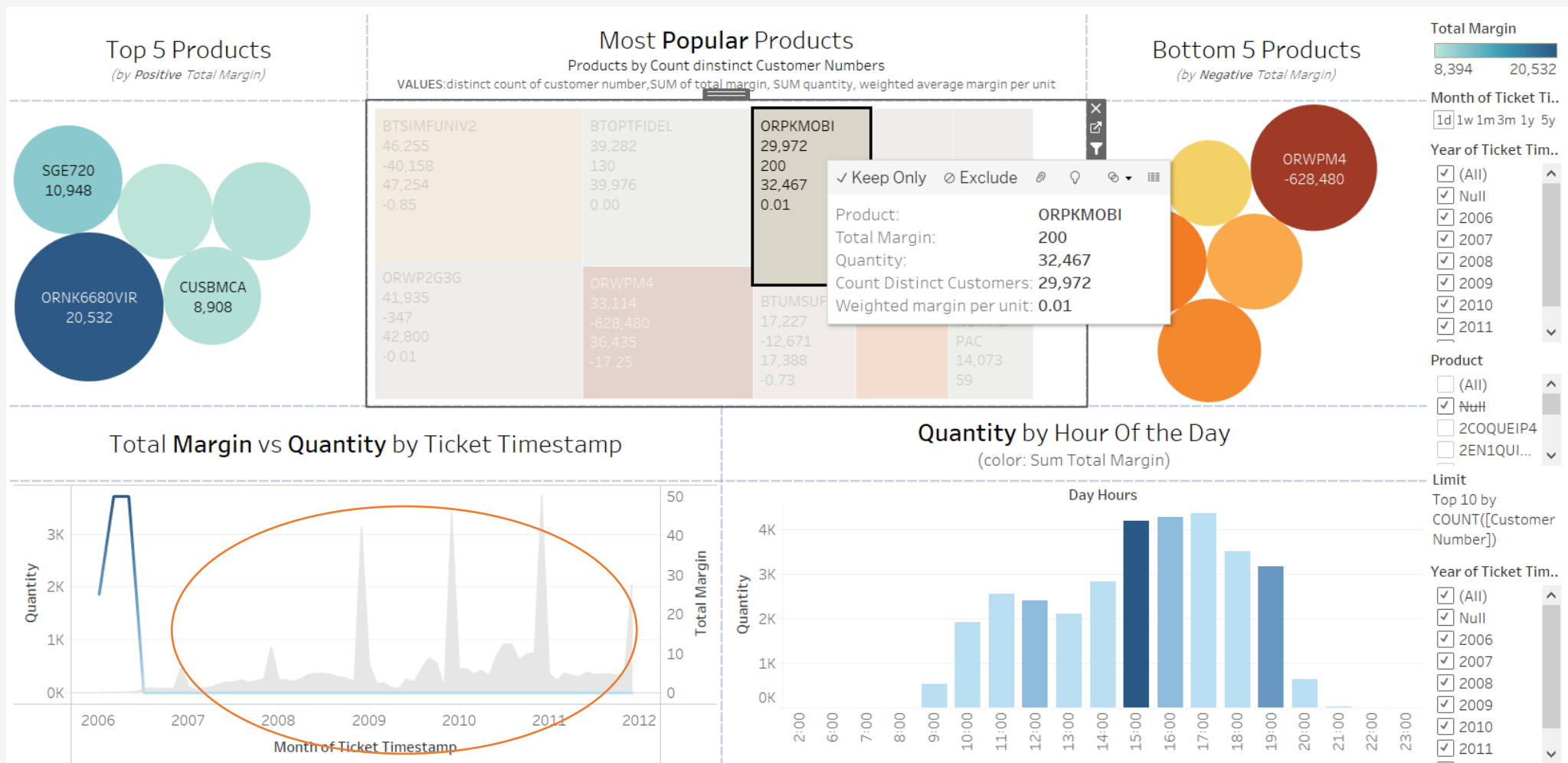
Between the most popular products there are products with zero margin but with quantities sold, which means that even though these products are not generating profit, they are still being sold in some quantity.

Other Example in the next slide



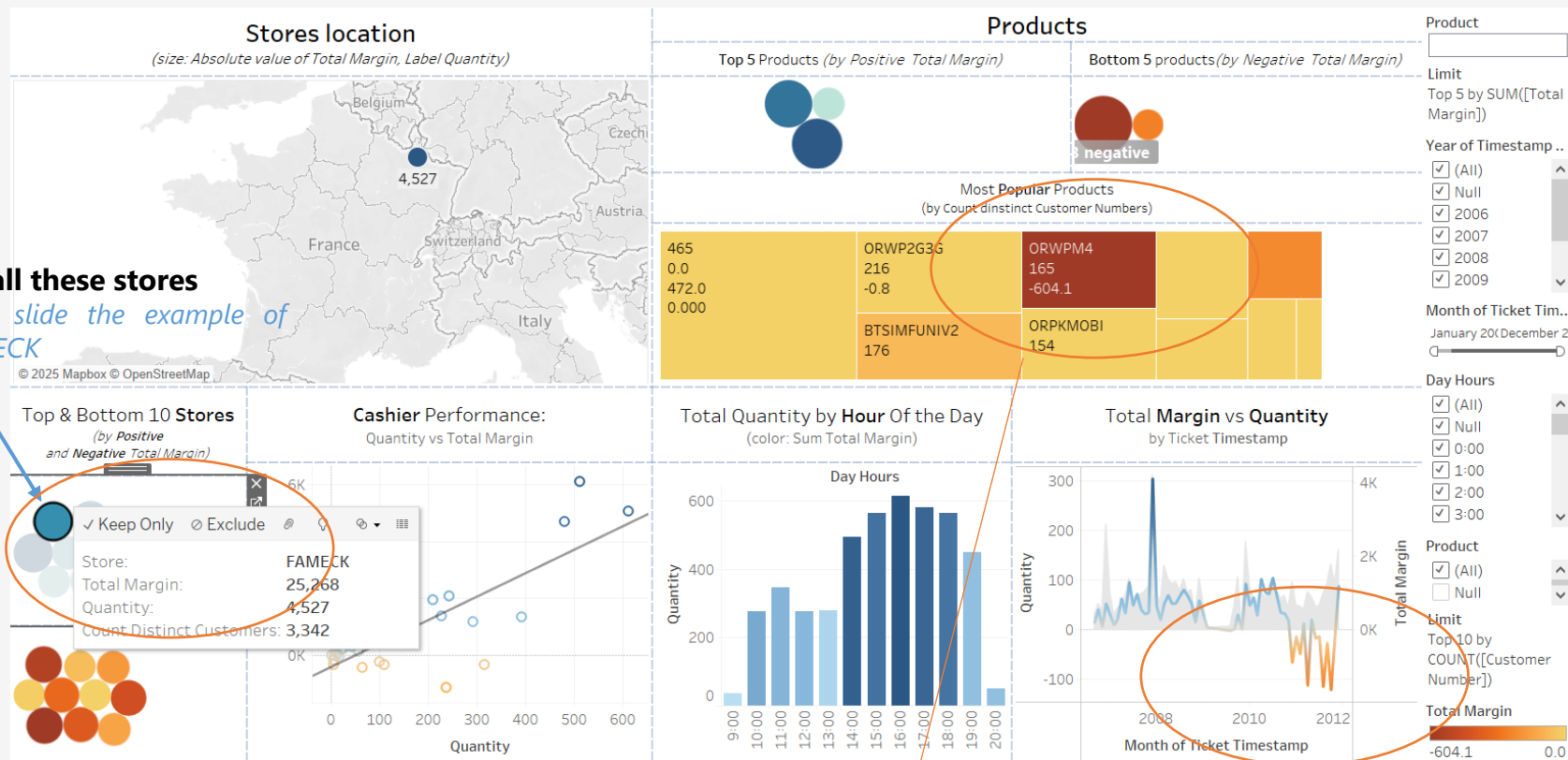
- **Quantity positive**
- **Margin 0**

5. Analysis of the Most Popular Products Among Customers



- Quantity positive
- Margin 0

5. Store Analysis: Identifying the Product Behind Margin Reduction

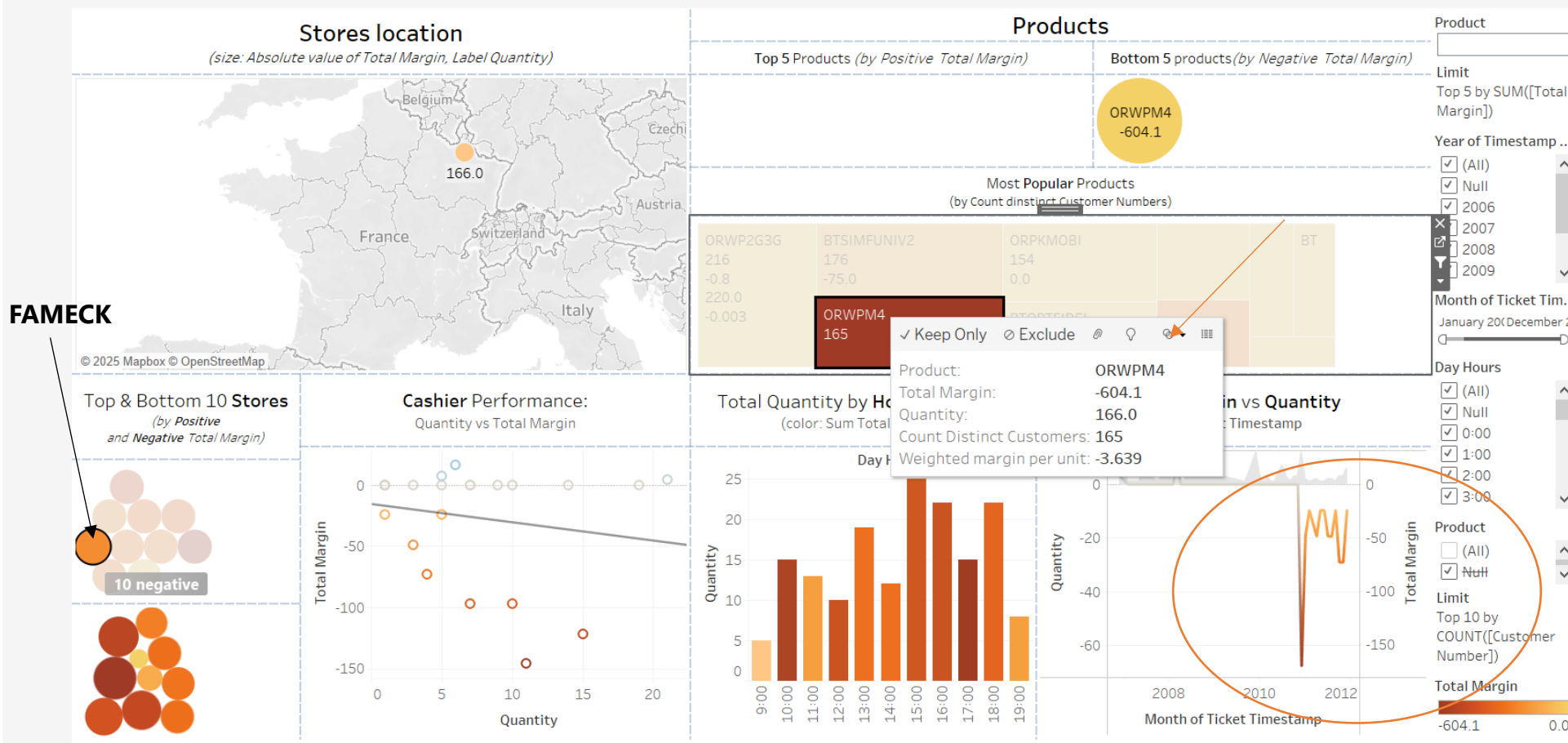


Considerations valid for **all** top 10 stores by positive total margin:

- After 2009/2010, most of these stores have recorded losses.
- The common product sold across these stores is ORWPM4 that has a significant negative impact on the margin, with high quantities sold and a negative weighted margin per unit.
- Filtering by the product (in the next slide), shows a clear influence on the negative trend in sales and margin across the store.



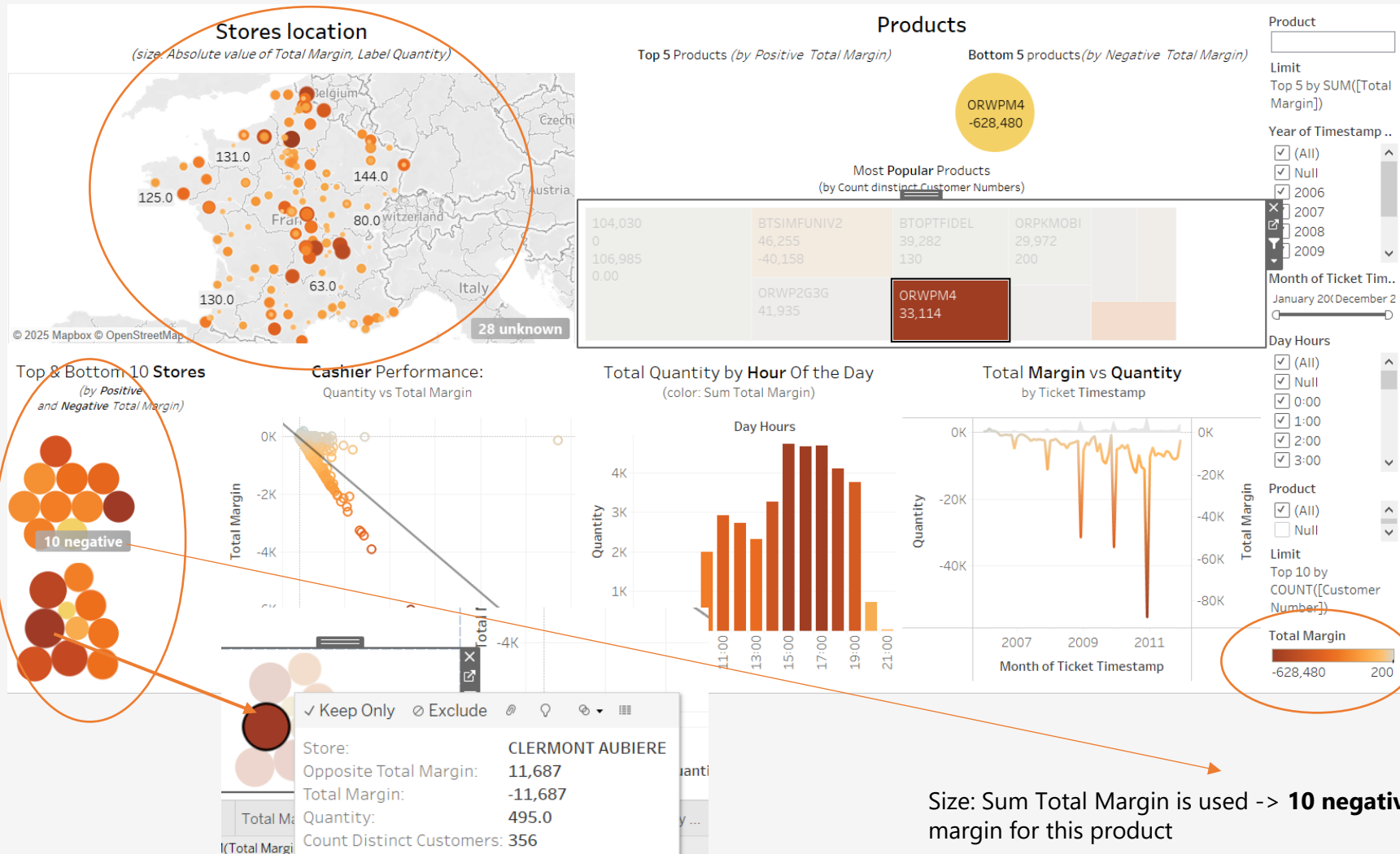
5. Example of the impact of ORPWPM4 in a Store



Filtered by both store (**FAMECK**) and product (**ORWPM4**):

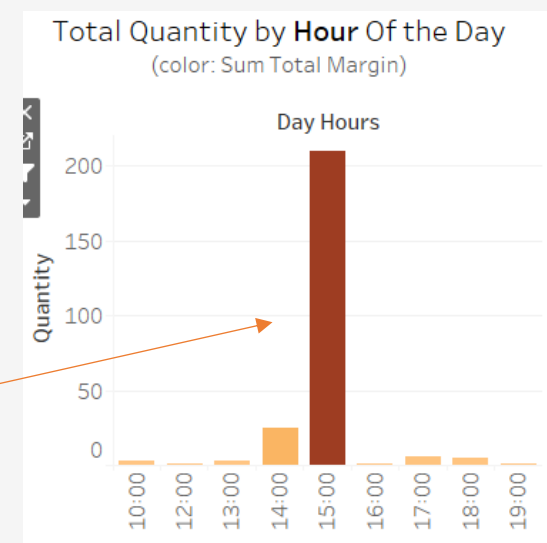
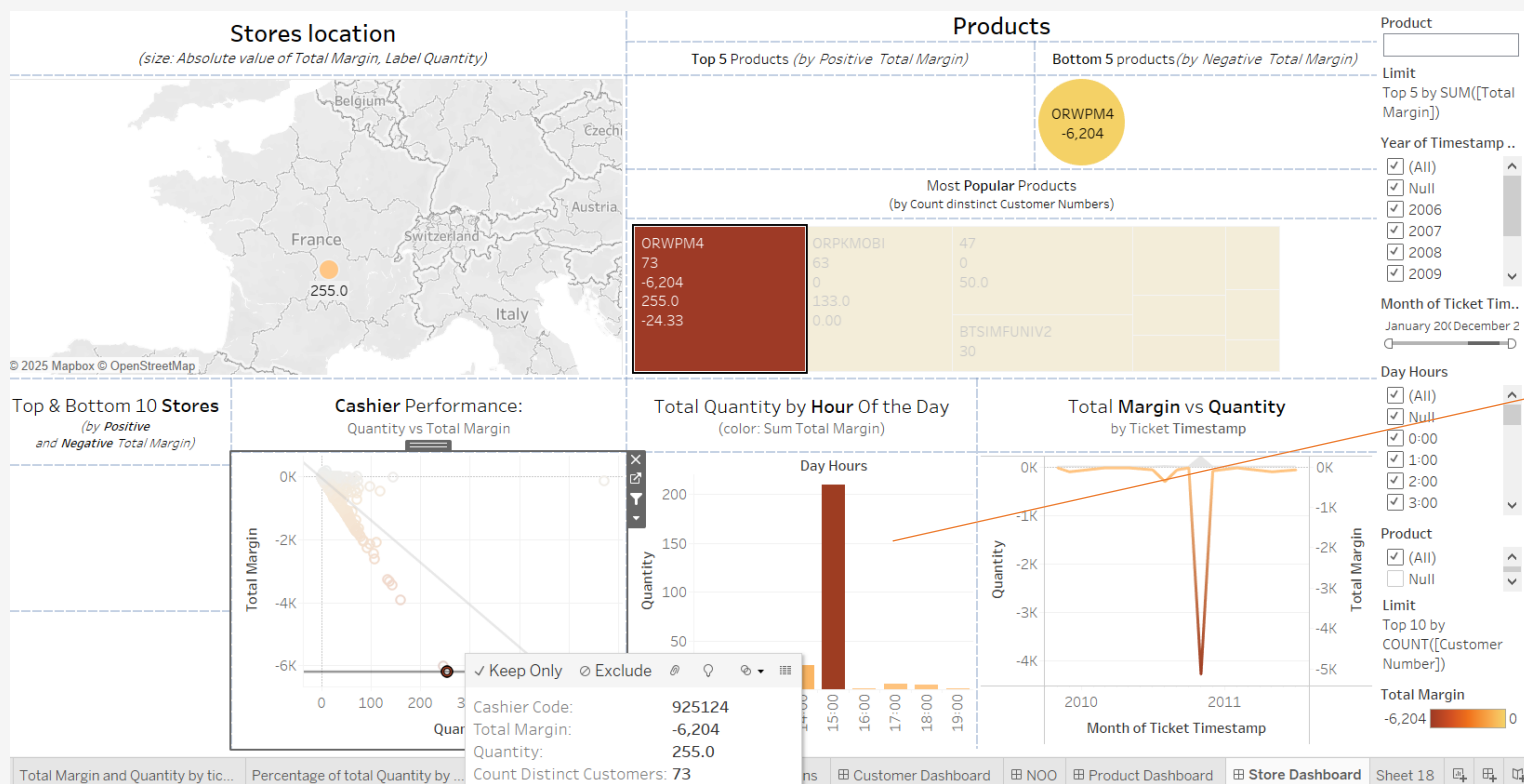
- Quantity sold of this product in the store: 166 units
- Weighted margin per unit in the store for this product: -3.639
- Purchased by 165 distinct customers

5. ORPWM4 across the stores



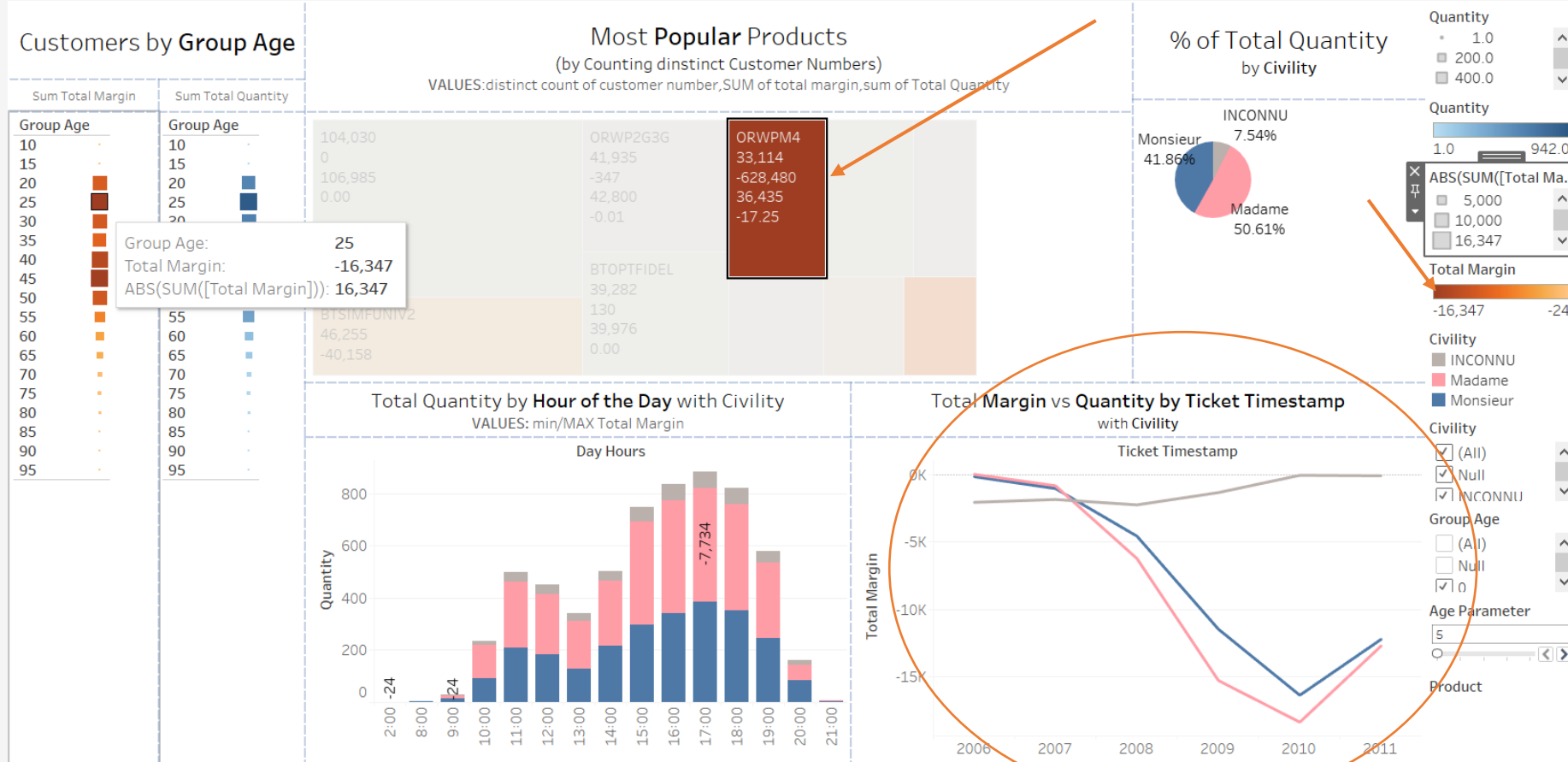
- While first filtering by the product and then by the store, it can be observed that among the stores with both most positive and most negative margins for the product, all margins are negative (all red).
- The lowest negative margin is from the store Clermont Aubière
- From the map it can be observed how the product is widely distributed

5. And the cashiers?



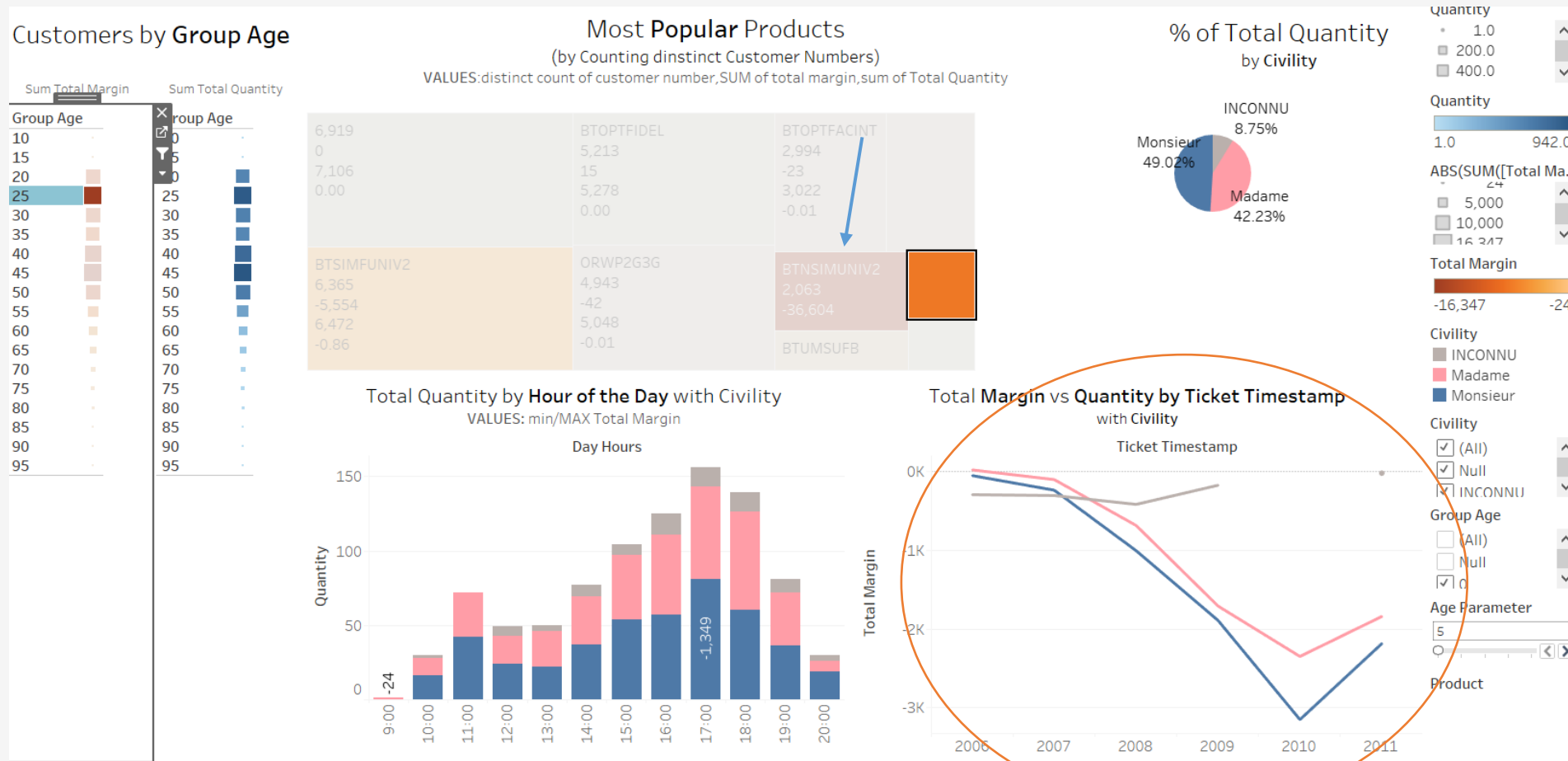
It can be observed that the cashier who generated the most negative margin for this product mainly sold it at 3 PM.

5. ORWPM4 by customers



- The 25 to 30 age group generated the highest total negative margin of the product
Even though, in terms of quantity, this product was bought more by the 45 to 50 age group.
- Women have had a greater impact on the negative margin of the product

5. ORWPM4 by 25-30 group Age

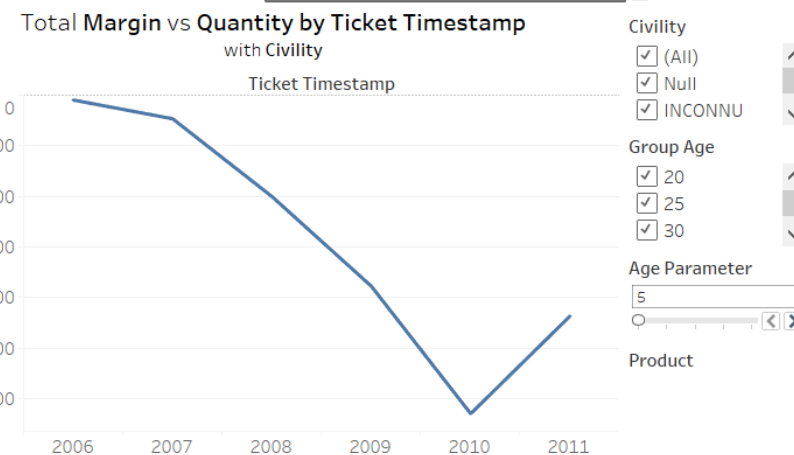
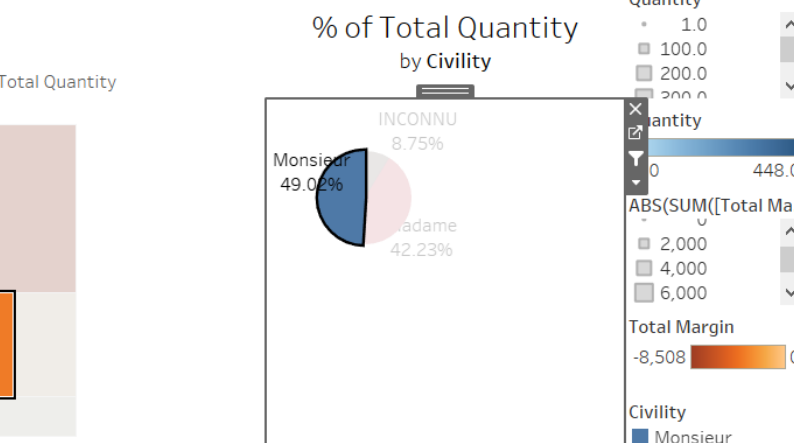
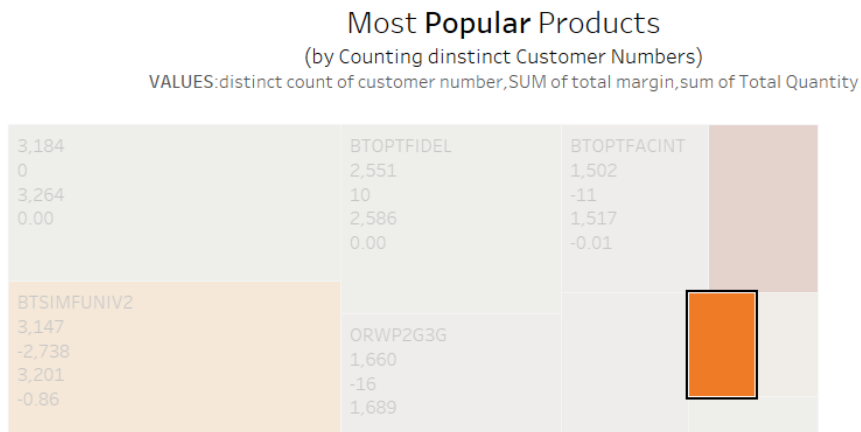


However, when filtering by both product and the 25-30 age group, it is the males who have contributed the most to the negative margin in this age range.

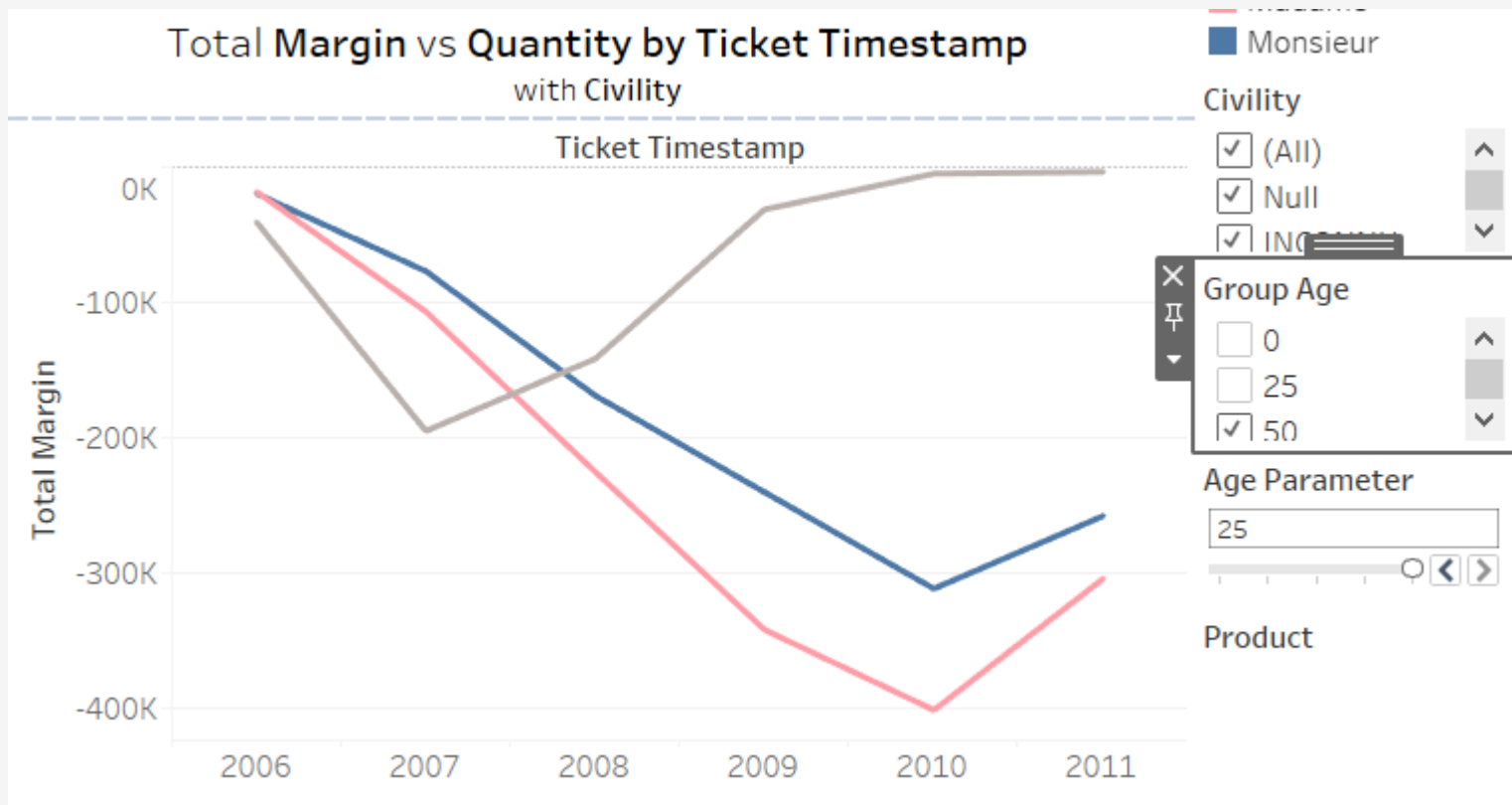
(Additionally, it is evident that another product has generated a greater negative margin in this age group.)

In the next slide, it can be seen how the male customers in the 25-30 age group who purchased ORWPM4 have influenced the margin.

Customers by Group Age

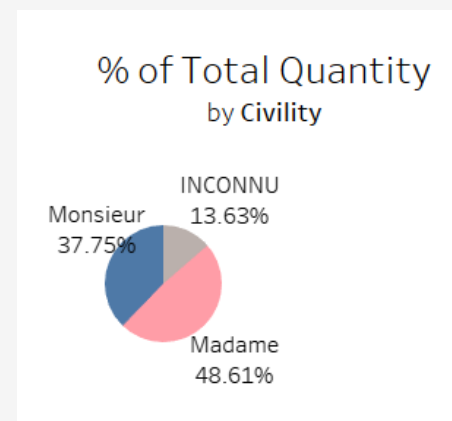


5. Age Group across all products



In general women and men have similar behaviour but considering the age group **>=50**

1. women contribute the most to the negative sales margin
2. purchasing a higher percentage of products compared to men



6. Conclusion

The analysis focused on identifying the causes of margin reduction.

- First, **products with the highest positive total margin** were examined, with sales limited to 2006 and low quantities sold, showing only a high weighted margin per unit.
- Next, the most purchased **products by distinct customers** were analyzed, revealing some with a zero margin despite positive quantities sold.
- The analysis shifted to **stores**, investigating whether a common product was contributing to the margin decrease. Once one of the potential product was identified, the store with the lowest negative margin for that product was found, and the cashier responsible for the lowest margin at that store was analyzed.
- Then, the focus moved to the customers of **that product**, exploring correlations with **Civility** and **Age Group**. It was found that women contributed the most to the lowest margin of the product, and the 25-30 age group had the lowest margin (of the product). However, when analyzing the 25-30 age group for this product, it was the men who caused the lowest margin.

This analysis highlights the key factors across products, stores, and customers that could have contributed to the reduction in margin over time